

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2013-14

Paper: BSC-044 — Intro To Computer Programming C

Time: Three Hours

Maximum Marks: 70

Instructions: Answer any five questions, including Question No. 1, which is compulsory. All undefined symbols have their usual meanings.

Question 1. Answer any six questions:

[3 × 6 = 18]

- (a) What are the steps involved in solving problems using computer?
- (b) What are logical operators? List them with their meaning.
- (c) What is the difference between `scanf()` and `printf()`?
- (d) What is bottom up approach of program designing?
- (e) Define data type. Write the names of any three data types.
- (f) What is a header file? Write the names of any three header files.
- (g) What is the difference between local and global variables?

Question 2.

- (a) What is an algorithm? Draw the flow chart for comparing three given numbers and display the largest number. [5]
- (b) What are the advantages of High level languages over Machine level languages? Write the names of any three high level languages. [3]
- (c) What are the differences between Compiler and Interpreter? [3]

Question 3.

- (a) Explain the various decision making statements in C language with an example for each. [6]
- (b) Explain the difference between a 'while' loop and a 'do-while' loop with a suitable example. [6]

Question 4.

- (a) Discuss about the precedence of operators in C. What will be the output of the following program: [4]

```
void main()
{
    int x;
    x = -3 + 4 - 7 * 8 / 5 % 10;
```

```
printf("x=%d\n", x);  
}
```

- (b) What is an array? How do you declare and initialize a one dimensional array? Illustrate with an example. [5]
- (c) Write a 'C' program to convert the given temperature in Fahrenheit to Celsius. [4]

Question 5.

- (a) Define pointer. What are the advantages of using pointers? [4]
- (b) What is the difference between a String and Array of characters? [3]
- (c) Explain any three String handling library functions in 'C' with an example for each. [6]

Question 6.

- (a) Define Function in C language. What is the difference between a user defined function and library function? [4]
- (b) Explain the parameter passing techniques of functions through an example. [6]
- (c) What do you mean by function prototype? [3]

Question 7.

- (a) Define storage classes in C Programming. List various storage classes used in 'C' with their properties. [6]
- (b) What is a structure? How do you declare a structure variable and access its members? Explain with an example. [7]